



# Full-Twist Presentations for Fundamental Groups of Complements of Complexified Hyperplane Arrangements

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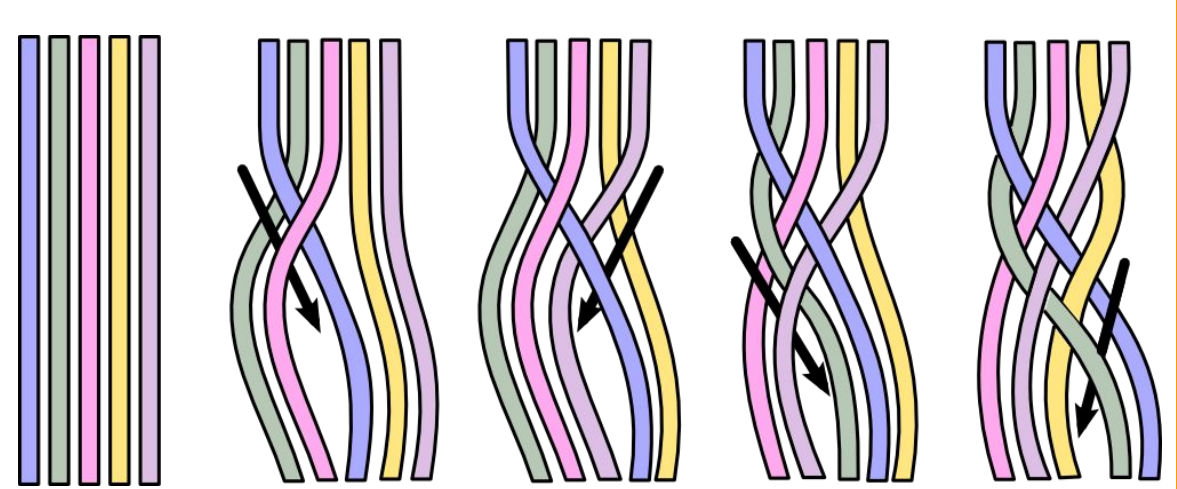
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## "Violent Protest and Discrimination"

Artin proved in 1926 that the *braid group*  $B_n$  has the presentation (think symmetric group and simple transpositions)

$$B_n = \langle s_1, \dots, s_{n-1} : \begin{array}{ll} s_i s_j = s_j s_i & \text{if } |i - j| > 1 \\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} & \text{for } 2 \leq i+1 \leq n-1 \end{array} \rangle.$$



Although it has been proved that every braid can be deformed into a similar normal form the writer is convinced that any attempt to carry this out on a living person would only lead to **violent protests and discrimination** against mathematics - Artin

## Introduction

Fix a finite Coxeter group  $W$  with real (central, simplicial) hyperplane arrangement  $\mathcal{H}$  in  $\mathbb{R}^n$ . Write  $V_{\mathbb{C}}^{\text{reg}} := \mathbb{C}^n \setminus \mathcal{H}_{\mathbb{C}}$ .

### DEFINITION

The *braid group* of  $W$  is defined to be  $B(W) := \pi_1(V_{\mathbb{C}}^{\text{reg}}/W)$ .

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The *pure braid group* of type  $W$  is defined to be

$$P(W) = \pi_1(V_{\mathbb{C}}^{\text{reg}}).$$

The pure braid group fits into the exact sequence

$$1 \rightarrow P(W) \rightarrow B(W) \rightarrow W \rightarrow 1$$

and is generated by  $\mathbb{T}$ , the squares of the dual generators of  $B(W)$

Write  $\mathbb{C}$  for the *full twist*—the generator of the center of  $P(W)$ .

Let  $\text{Red}_{\mathbb{T}}(\mathbb{C})$  be the set of reduced words in  $\mathbb{T}$  for  $\mathbb{C}$ .

### Main Conjecture

$$P(W) = \langle \mathbb{T} : [\text{Red}_{\mathbb{T}}(\mathbb{C})] \rangle.$$

## Future Works and Acknowledgement

- Complete rewriting process for the Coxeter group of type  $F_4$  and extend the methods to the remaining exceptional groups. Beyond Coxeter types, we will attempt to prove the conjecture for all simplicial hyperplane arrangements.
- Thanks to the MIT-PRIMES program for making this research opportunity possible. Thanks to my mentor Nathan Williams for his guidance and support.

## Salvetti Complex and Presentation

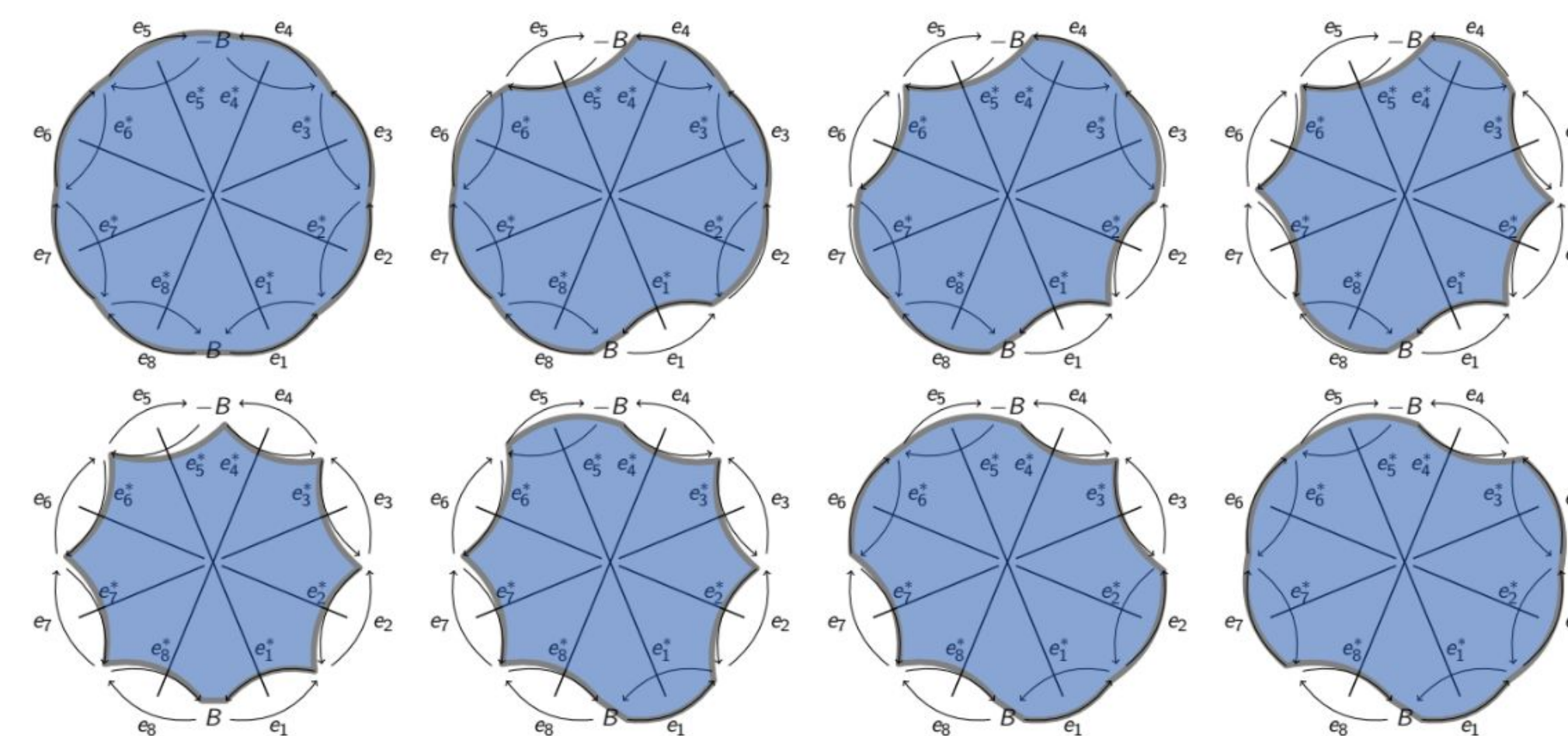
### COMBINATORIAL MODEL FOR $V_{\mathbb{C}}^{\text{reg}} = \mathbb{C}^n \setminus \mathcal{H}_{\mathbb{C}}$

Fix a base point  $x_B$  in the interior of some region  $B$ . We only care about  $\pi_1(V_{\mathbb{C}}^{\text{reg}}, x_B)$ , so only need 0, 1, and 2-cells.

**0-CELLS** Regions of the hyperplane arrangement.

**1-CELLS** For each pair of adjacent regions, edge  $e$  oriented away from  $B$  and an edge  $e^*$  oriented towards  $B$ .

**2-CELLS** Indexed by rank-two hyperplane intersections—glue one 2-cell for each region containing that intersection.



**Theorem 3.1** ([Sal87]). Let  $\mathcal{H}$  be a hyperplane arrangement, then

$$\pi_1(\mathbb{C}^n \setminus \mathcal{H}, x_B) \cong \pi_1(\mathcal{P}^*(\mathcal{H}, B), B).$$

**Theorem 3.2** ([Sal87], Lemma 12, Corollary 12). For any positive minimal gallery  $\mathbf{b}$  from  $B$  to  $-B$ ,  $\mathbb{T}_{\mathbf{b}}$  is a generating set of  $\pi_1(\mathcal{P}^*(\mathcal{H}, B), B)$ . Specifically, if  $C \xrightarrow{e} C'$  with  $H_e = H_{b_k}$ , then

$$\mathbb{t}_e = \left( \prod_{k \geq i \geq 1, H_i \notin S(e)} \mathbb{t}_{b_i} \right)^{-1} \mathbb{t}_{b_k} \left( \prod_{k \geq i \geq 1, H_i \notin S(e)} \mathbb{t}_{b_i} \right).$$

**Theorem 3.3** ([Sal87], Page 616). Fix a positive minimal gallery  $\mathbf{b}$  from  $B$  to  $-B$  in  $\mathcal{P}^*(\mathcal{H}, B)$ . For each full rank two subarrangement  $\mathcal{A}$  of  $\mathcal{H}$ , we can choose one two-cell in  $\mathcal{P}^*(\mathcal{H}, B)$  with edges labeled

$$e_1, e_2, \dots, e_M, \quad e_{M+1}, \dots, e_{2M}$$

where  $\{H_{e_i}\}_{i=1}^M = \mathcal{A}$  and  $H_{e_1} <_{\mathbf{b}} \dots <_{\mathbf{b}} H_{e_M}$ . Denote  $[\mathcal{A}]_{T_{\mathbf{b}}}$  as representing the relations

$$\mathbb{t}_{e_M} \dots \mathbb{t}_{e_2} \mathbb{t}_{e_1} = \mathbb{t}_{e_1} \mathbb{t}_{e_M} \dots \mathbb{t}_{e_2} = \dots = \mathbb{t}_{e_{M-1}} \dots \mathbb{t}_{e_2} \mathbb{t}_{e_1} \mathbb{t}_{e_M}.$$

Then

$$\pi_1(\mathcal{P}^*(\mathcal{H}, B), B) = \langle \mathbb{T}_{\mathbf{b}} : [\mathcal{A}]_{T_{\mathbf{b}}} \rangle.$$

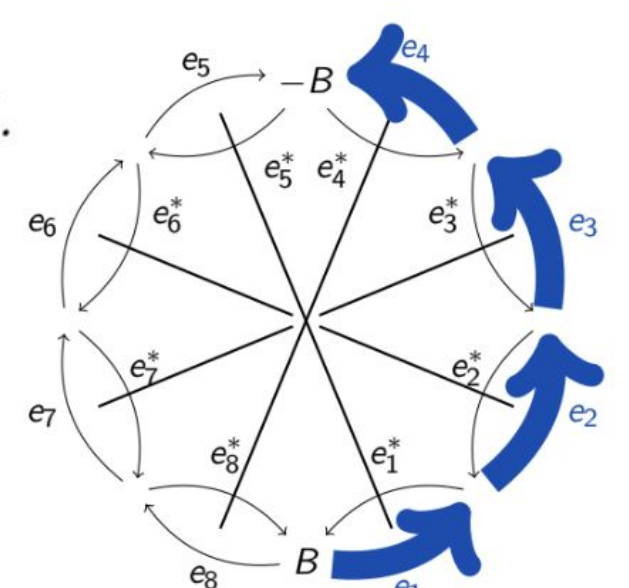
where the relations range over all full rank-two subarrangements  $\mathcal{A} \subset \mathcal{H}$ .

A positive minimal gallery

$$\mathbf{b} := b_1, b_2, \dots, b_N$$

from  $B$  to  $-B$  specifies a set  $\mathbb{T}_{\mathbf{b}} = \{\mathbb{t}_{b_i}\}_{i=1}^N$  by the formula:

$$\mathbb{t}_{b_i} := (b_1 b_2 \dots b_{i-1}) b_i b_i^* (b_{i-1}^{-1} \dots b_2^{-1} b_1^{-1}) \text{ for } 1 \leq i \leq N,$$



## Example Computation for Coxeter Group of Type A

| Type $A_{n-1}$ $c$ -Noncrossing Full Rank-Two Subarrangements |                                          |                               |
|---------------------------------------------------------------|------------------------------------------|-------------------------------|
| Label                                                         | $\mathcal{N}_c(A_{n-1}, A_1 \times A_1)$ | $\mathcal{N}_c(A_{n-1}, A_2)$ |
| Subarrangement                                                | $\{H_{ij}, H_{rs}\}$                     | $\{H_{ij}, H_{ik}, H_{jk}\}$  |
| Conditions                                                    | $i < j < r < s$ or $i < r < s < j$       | $i < j < k$                   |
| Picture                                                       |                                          |                               |
| Relation                                                      | $[(ij)(rs)]$                             | $[(ij)(ik)(jk)]$              |
| Type $A_{n-1}$ $c$ -Crossing Full Rank-Two Subarrangements    |                                          |                               |
| Label                                                         | $\mathcal{C}_c(A_{n-1}, A_1 \times A_1)$ |                               |
| Subarrangement                                                | $\{H_{ij}, H_{rs}\}$                     |                               |
| Conditions                                                    | $i < r < j < s$                          |                               |
| Picture                                                       |                                          |                               |
| Relation                                                      | $[(ij)(rs)(js)]$                         |                               |



FIGURE 1. The full rank-two subarrangements of type  $A_{n-1}$ .

## THEOREM (ARTIN, MARGALIT-McCAMMOND, W.)

Let  $c = s_1 s_2 \dots s_{n-1}$ . Then

$$P_n = \left\langle \mathbb{T}_c \left| \begin{array}{ll} [(ij)(rs)] & \text{if } \{H_{ij}, H_{rs}\} \in \mathcal{N}_c(A_{n-1}, A_1 \times A_1) \\ [(ij)(ik)(jk)] & \text{if } \{H_{ij}, H_{ik}, H_{jk}\} \in \mathcal{N}_c(A_{n-1}, A_2) \\ [(ij)(rs)(js)] & \text{if } \{H_{ij}, H_{rs}\} \in \mathcal{C}_c(A_{n-1}, A_1 \times A_1) \end{array} \right. \right\rangle$$

**Theorem 6.4.** The pure braid group  $P(A_{n-1})$  has the positive presentation:

$$P(A_{n-1}) = \left\langle \mathbb{T}_c \left| \begin{array}{ll} [(ij)(rs)] & \text{if } \{H_{ij}, H_{rs}\} \in \mathcal{N}_c(A_{n-1}, A_1 \times A_1) \\ [(ij)(ik)(jk)] & \text{if } \{H_{ij}, H_{ik}, H_{jk}\} \in \mathcal{N}_c(A_{n-1}, A_2) \\ [(ij)(is)(rs)(js) = (is)(rs)(js)(ij)] & \text{if } \{H_{ij}, H_{rs}\} \in \mathcal{C}_c(A_{n-1}, A_1 \times A_1) \end{array} \right. \right\rangle.$$

*Proof.* We only have to confirm that the positive  $c$ -crossing relations are equivalent to the original  $c$ -crossing relations. This follows easily from the computation:

$$\begin{aligned} (ij)(js)(rs)(js) &= (js)(rs)(js)(ij) \\ (js)(ij)(js)(rs)(js) &= (rs)(js)(ij) \\ (is)(js)(ij)(js)(rs)(js) &= (is)(rs)(js)(ij) \\ (ij)(is)(rs)(js) &= (is)(rs)(js)(ij). \end{aligned}$$

The last step invokes a relation of type  $\mathcal{N}_c(A_{n-1}, A_2)$ .  $\square$